4

2

0

2

4

432101234 Avg. RTE: 0.0101

Num Aligned: 9

Ground Truth bonsai: COLMAP | | num_images=20 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

42024

Avg. RTE: 0.5028
Num Aligned: 14

Ground Truth bonsai: COLMAP | | num_images=30 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

432101234 Avg. RTE: 0.0130

Num Aligned: 26

Ground Truth bonsai: COLMAP | | num_images=40 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

432101234 Avg. RTE: 0.0132

Num Aligned: 100

Ground Truth bonsai: COLMAP | | num_images=100 10 3 10 2 10 1 100 Translation Error (RTE)
4

2

0

2

4

42024 Avg. RTE: 0.1056

Num Aligned: 20

Ground Truth bonsai: VGGT | | num_images=20 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

42024 Avg. RTE: 0.1097

Num Aligned: 30

Ground Truth bonsai: VGGT | | num_images=30 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

42024 Avg. RTE: 0.1038

Num Aligned: 40

Ground Truth bonsai: VGGT | | num_images=40 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

42024 Avg. RTE: 0.1057

Num Aligned: 100

Ground Truth bonsai: VGGT | | num_images=100 10 3 10 2 10 1 100

Translation Error (RTE)

4

2

0

2

4

432101234 Avg. RTE: 0.0093

Num Aligned: 12

Ground Truth kitchen: COLMAP | | num_images=20 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

432101234 Avg. RTE: 0.0082

Num Aligned: 19

Ground Truth kitchen: COLMAP | | num_images=30 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

432101234 Avg. RTE: 0.0073

Num Aligned: 40

Ground Truth kitchen: COLMAP | | num_images=40 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

432101234 Avg. RTE: 0.0045

Num Aligned: 100

Ground Truth kitchen: COLMAP | | num_images=100 10 3 10 2 10 1 100 Translation Error (RTE)
4

2

0

2

4

432101234 Avg. RTE: 0.0293

Num Aligned: 20

Ground Truth kitchen: VGGT | | num_images=20 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

432101234 Avg. RTE: 0.0281

Num Aligned: 30

Ground Truth kitchen: VGGT | | num_images=30 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

432101234 Avg. RTE: 0.0331

Num Aligned: 40

Ground Truth kitchen: VGGT | | num_images=40 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

432101234 Avg. RTE: 0.0310

Num Aligned: 100

Ground Truth kitchen: VGGT | | num_images=100 10 3 10 2 10 1 100

Translation Error (RTE)

4

2

0

2

4

42024

Avg. RTE: 0.0010
Num Aligned: 3

Ground Truth counter: COLMAP | | num_images=20 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

42024

Avg. RTE: 0.0325
Num Aligned: 14

Ground Truth counter: COLMAP | | num_images=30 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

42024

Avg. RTE: 0.0050
Num Aligned: 22

Ground Truth counter: COLMAP | | num_images=40 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

42024

Avg. RTE: 0.0021
Num Aligned: 100

Ground Truth counter: COLMAP | | num_images=100 10 3 10 2 10 1 100 Translation Error (RTE)
4

2

0

2

4

42024

Avg. RTE: 0.0364
Num Aligned: 20

Ground Truth counter: VGGT | | num_images=20 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

42024

Avg. RTE: 0.0355
Num Aligned: 30

Ground Truth counter: VGGT | | num_images=30 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

42024

Avg. RTE: 0.0407
Num Aligned: 40

Ground Truth counter: VGGT | | num_images=40 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

42024

Avg. RTE: 0.0329
Num Aligned: 100

Ground Truth counter: VGGT | | num_images=100 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

42024 Avg. RTE: 0.0000

Num Aligned: 2

Ground Truth bicycle: COLMAP | | num_images=20 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

42024 Avg. RTE: 0.0100

Num Aligned: 8

Ground Truth bicycle: COLMAP | | num_images=30 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

42024 Avg. RTE: 0.0203

Num Aligned: 11

Ground Truth bicycle: COLMAP | | num_images=40 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

42024 Avg. RTE: 0.0069

Num Aligned: 100

Ground Truth bicycle: COLMAP | | num_images=100 10 3 10 2 10 1 100 Translation Error (RTE)
4

2

0

2

4

42024 Avg. RTE: 0.1240

Num Aligned: 20

Ground Truth bicycle: VGGT | | num_images=20 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

42024 Avg. RTE: 0.0916

Num Aligned: 30

Ground Truth bicycle: VGGT | | num_images=30 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

42024 Avg. RTE: 0.0894

Num Aligned: 40

Ground Truth bicycle: VGGT | | num_images=40 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

42024 Avg. RTE: 0.0656

Num Aligned: 100

Ground Truth bicycle: VGGT | | num_images=100 10 3 10 2 10 1 100

Translation Error (RTE)

4

2

0

2

4

432101234

Avg. RTE: 0.0047
Num Aligned: 20

Ground Truth garden: COLMAP | | num_images=20 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

432101234

Avg. RTE: 0.0035
Num Aligned: 30

Ground Truth garden: COLMAP | | num_images=30 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

432101234

Avg. RTE: 0.0032
Num Aligned: 40

Ground Truth garden: COLMAP | | num_images=40 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

432101234

Avg. RTE: 0.0026
Num Aligned: 100

Ground Truth garden: COLMAP | | num_images=100 10 3 10 2 10 1 100 Translation Error (RTE)
4

2

0

2

4

432101234

Avg. RTE: 0.0214
Num Aligned: 20

Ground Truth garden: VGGT | | num_images=20 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

432101234

Avg. RTE: 0.0200
Num Aligned: 30

Ground Truth garden: VGGT | | num_images=30 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

432101234

Avg. RTE: 0.0248
Num Aligned: 40

Ground Truth garden: VGGT | | num_images=40 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

432101234

Avg. RTE: 0.0269
Num Aligned: 100

Ground Truth garden: VGGT | | num_images=100 10 3 10 2 10 1 100

Translation Error (RTE)

4

2

0

2

4

432101234 Avg. RTE: 0.0309

Num Aligned: 6

Ground Truth stump: COLMAP | | num_images=20 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

432101234 Avg. RTE: 0.0036

Num Aligned: 11

Ground Truth stump: COLMAP | | num_images=40 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

432101234 Avg. RTE: 0.0020

Num Aligned: 100

Ground Truth stump: COLMAP | | num_images=100 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

432101234 Avg. RTE: 0.2988

Num Aligned: 20

Ground Truth stump: VGGT | | num_images=20 10 3 10 2 10 1 100

Translation Error (RTE)

4

2

0

2

4

432101234 Avg. RTE: 0.0000

Num Aligned: 2

Ground Truth stump: COLMAP | | num_images=30 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

432101234

Avg. RTE: 0.1496
Num Aligned: 30

Ground Truth stump: VGGT | | num_images=30 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

432101234

Avg. RTE: 0.1259
Num Aligned: 40

Ground Truth stump: VGGT | | num_images=40 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

32101234

Avg. RTE: 0.1628
Num Aligned: 100

Ground Truth stump: VGGT | | num_images=100 10 3 10 2 10 1 100

Translation Error (RTE)

4

2

0

2

4

432101234

Avg. RTE: 2.4757
Num Aligned: 10

Ground Truth lego: COLMAP | | num_images=20 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

432101234

Avg. RTE: 3.7453
Num Aligned: 19

Ground Truth lego: COLMAP | | num_images=30 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

432101234

Avg. RTE: 3.7605
Num Aligned: 39

Ground Truth lego: COLMAP | | num_images=40 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

432101234

Avg. RTE: 0.4038
Num Aligned: 87

Ground Truth lego: COLMAP | | num_images=87 10 3 10 2 10 1 100

Translation Error (RTE)

4

2

0

2

4

432101234

Avg. RTE: 0.1366
Num Aligned: 20

Ground Truth lego: VGGT | | num_images=20 10

3

1

0

2

1

0

1

1

0

0

Translation Error (RTE)
4

2

0

2

4

432101234

Avg. RTE: 0.1312
Num Aligned: 30

Ground Truth lego: VGGT | | num_images=30 10

3

1

0

2

1

0

1

1

0

0

Translation Error (RTE)
4

2

0

2

4

432101234

Avg. RTE: 0.0689
Num Aligned: 40

Ground Truth lego: VGGT | | num_images=40 10

3

1

0

2

1

0

1

1

0

0

Translation Error (RTE)
4

2

0

2

4

432101234

Avg. RTE: 0.0188
Num Aligned: 87

Ground Truth lego: VGGT | | num_images=87 10

3

1

0

2

1

0

1

1

0

0

Translation Error (RTE)

4

2

0

2

4

432101234

Avg. RTE: 0.0115
Num Aligned: 5

Ground Truth ship: COLMAP | | num_images=20 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

432101234

Avg. RTE: 0.0430
Num Aligned: 10

Ground Truth ship: COLMAP | | num_images=30 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

432101234

Avg. RTE: 0.0169
Num Aligned: 12

Ground Truth ship: COLMAP | | num_images=40 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

432101234

Avg. RTE: 0.0112
Num Aligned: 80

Ground Truth ship: COLMAP | | num_images=87 10 3 10 2 10 1 100

Translation Error (RTE)

4

2

0

2

4

432101234

Avg. RTE: 0.0230
Num Aligned: 20

Ground Truth ship: VGGT | | num_images=20 10

3

1

0

2

1

0

1

1

0

0

Translation Error (RTE)
4

2

0

2

4

432101234

Avg. RTE: 0.0195
Num Aligned: 30

Ground Truth ship: VGGT | | num_images=30 10

3

1

0

2

1

0

1

1

0

0

Translation Error (RTE)
4

2

0

2

4

432101234

Avg. RTE: 0.0181
Num Aligned: 40

Ground Truth ship: VGGT | | num_images=40 10

3

1

0

2

1

0

1

1

0

0

Translation Error (RTE)
4

2

0

2

4

432101234

Avg. RTE: 0.0278
Num Aligned: 87

Ground Truth ship: VGGT | | num_images=87 10

3

1

0

2

1

0

1

1

0

0

Translation Error (RTE)

4

2

0

2

4

432101234 Avg. RTE: 0.0038

Num Aligned: 3

Ground Truth drums: COLMAP | | num_images=20 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

432101234 Avg. RTE: 0.0117

Num Aligned: 4

Ground Truth drums: COLMAP | | num_images=30 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

432101234 Avg. RTE: 0.0056

Num Aligned: 5

Ground Truth drums: COLMAP | | num_images=40 10 3 10 2 10 1 100 Translation Error (RTE)

4

2

0

2

4

432101234 Avg. RTE: 0.0099

Num Aligned: 79

Ground Truth drums: COLMAP | | num_images=87 10 3 10 2 10 1 100 Translation Error (RTE)
4

2

0

2

4

432101234 Avg. RTE: 0.0526

Num Aligned: 20

Ground Truth drums: VGGT | | num_images=20 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

432101234 Avg. RTE: 0.0414

Num Aligned: 30

Ground Truth drums: VGGT | | num_images=30 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

432101234 Avg. RTE: 0.0194

Num Aligned: 40

Ground Truth drums: VGGT | | num_images=40 10 3 10 2 10 1 100

Translation Error (RTE)
4

2

0

2

4

432101234 Avg. RTE: 0.0100

Num Aligned: 87

Ground Truth drums: VGGT | | num_images=87 10 3 10 2 10 1 100

Translation Error (RTE)